

Amirali Tolooei

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EDUCATION

University of Toronto

Honours Bachelor of Science — Computer Science Specialist, Statistics Minor

September 2021 - August 2025

Toronto, ON

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C, C++, HTML/CSS, R, C#

Technologies/Frameworks: React, Next.js, Django, Flask, FastAPI, Tailwind CSS, Docker, PostgreSQL, SQLite, Prisma, Drizzle ORM, PyTorch, NumPy, Pandas, OpenAI API, ChromaDB, Puppeteer, Pytest, Vitest

Developer Tools: Git, GitHub, GitHub Actions, GCP (Cloud Run, Cloud SQL), AWS, Vercel, VS Code, CI/CD

Methodologies: Agile Development, REST API Design, Test-Driven Development, DevOps, Code Review

EXPERIENCE

Full-Stack Developer | HouseSigma

June 2026 - Present

Python, Twilio, Deepgram, AWS (Bedrock, S3), LLMs/VLMs, Kafka, GitHub Actions

Remote — Toronto, ON

- Built and deployed an AI voice agent that conducts real-time phone pre-screen interviews, using a **Twilio + Deepgram** speech pipeline with concurrent **LLM** classifiers for turn-taking and adaptive follow-ups.
- Built an **LLM-persona** test harness that exercises the live agent with synthetic candidates graded on a multi-axis rubric, and designed the real-time audio media bridge the calling system runs on.
- Designed and built an end-to-end **LLM evaluation platform** for a real-estate photo pipeline, benchmarking **15+ vision-language models** against a human-verified golden set with a validated **LLM-as-judge** cascade.
- Improved model output quality through iterative **prompt and context engineering**, and produced **inference cost analysis** to inform production-scale deployment.

Founding Engineer & Full-Stack Developer | Cypress

January 2026 - April 2026

Python, Django, React, TypeScript, Tailwind CSS, PostgreSQL, GCP, Docker

Remote — San Francisco, CA

- Joined the founding engineering team that took the platform from first revenue to **>US\$12M** in annualized GMV.
- Built a **Django REST Framework** API from scratch with JWT authentication, 3-tier role-based access control, invitation-based onboarding, and a configurable pricing policy engine integrating Cox Automotive's MMR API.
- Developed a **React 18 + TypeScript** SPA with shadcn/ui, implementing a Figma-to-code design system with role-conditional dashboards, data tables with sort/search/pagination, and a multi-step signup wizard.
- Provisioned **GCP** staging infrastructure (Cloud Run, Cloud SQL, Artifact Registry) with multi-stage Docker builds, and enforced strict code quality with mypy, ruff, and 400+ automated tests across unit and integration suites.

AI Developer | Montagy AI

May 2025 - October 2025

Python, Flask, FastAPI, SQLite, ChromaDB, AWS, Docker, GitHub Actions

Toronto, ON

- Deployed zero-shot TTS system using a diffusion transformer model in **Python Flask** application that clones user voices from short reference clips to generate personalized speech synthesis.
- Optimized production model pipelines using **FastAPI** deployment initially on Modal then migrated to **AWS** implementing server-level optimizations and caching strategies to achieve 98% uptime and reduce latency by 40%.
- Containerized applications using **Docker** and established CI/CD pipelines with **GitHub Actions**, automating testing, linting, and deployment workflows to streamline production releases and reduce manual deployment time.

PROJECTS

TableTally - MTG Companion | Next.js, TypeScript, Prisma, Turso, Puppeteer

January 2026 - Present

- Built a **Next.js 16 + React 19** app for an MTG playgroup that matches decklists to members' Moxfield collections, searches 4 local game stores' live inventory, and tracks game stats.
- Designed a concurrent scraping pipeline on serverless Chromium with **Puppeteer**, and replaced a Cloudflare-blocked headless scrape with direct **Shopify** JSON API calls using bounded concurrency and per-request timeouts.
- Implemented HMAC-signed session auth with the **Web Crypto API**, a route-scoped sliding-window rate limiter, and **Vercel Cron** collection syncs over a **Prisma/Turso** schema, backed by **689 Jest tests**.

Squeebot - Discord Chatbot | TypeScript, Node.js, SQLite, Modal, vLLM, LoRA

April 2026 - May 2026

- Built a **TypeScript** Discord bot with a 3-tier LLM cascade (Groq → Gemini → self-hosted fine-tune) and local **RAG** retrieval over 10K embeddings via **SQLite**, with per-user persistent memory.
- Fine-tuned Llama 3.1 8B via **Unsloth QLoRA** on Modal serverless GPU, served the AWQ-quantized model with **vLLM**, and deployed under **systemd** with crash-loop guards.